A. J. A. Jubayer Talukder

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Education

Bangladesh University of Engineering and Technology (BUET) — **Dhaka, Bangladesh**
Bachelor of Science in Electrical and Electronics Engineering (EEE) — *Jan. 2022 – June 2026*
Major in Communication and Signal Processing (CSP) — **CGPA: 3.83** (Out of 4.00)

Thesis and Ongoing Research

Proactive Conversation Assistant: LLM Pipeline | *NLP, Edge AI* — **Feb. 2025 – Present**

- Reimplemented a real-time wearable assistant that decides *when to interrupt* a conversation and *what to say*, using a two-stage LLM pipeline that substantially cuts unnecessary generator calls and latency.
- Found persistent user memory to be the strongest factor in hint quality; designed a compact decoder that shares one adapter for interruption detection and response generation, lowering memory and compute over the two-model pipeline.

Deepfake Detection using CLIP-ViT | *OpenCLIP, ViT, PyTorch* — **Oct. 2024 – Jan. 2025**

- Developed a deepfake detection model using OpenCLIP-ViT; achieved 98% training and 97% validation accuracy.
- Tested on IEEE SP Cup 2025 dataset (43k real, 219k fake images) with strict mini-batch training.

Publications

[1] **A. J. A. Jubayer Talukder** and M. A. Haque, "When to Assist and What to Suggest: Joint Gating and Memory-Grounded Hint Generation in One Compact Decoder," *arXiv preprint arXiv:2610.40611*, 2026.

[2] M. Nakib, S. Sarkar, **A. J. A. Jubayer Talukder**, and M. A. Haque, "PairCLIP-SWA: Similarity-Guided Balancing and Stochastic Weight Averaging for Generalized Deepfake Image Detection with OpenCLIP," *arXiv preprint arXiv:2610.37874*, 2026.

Projects

Autonomous Rescue Drone for Locating Survivors | *Computer Vision, AI, UAV* — **Sep. – Dec. 2024**

- Developed an AI-powered drone for search and rescue; implemented real-time human detection using RGB cameras and OpenCV.
- Integrated GPS-based autonomous flight control for structured area coverage.

Ultrasonic Obstacle Mapping Bot | *Embedded Systems, Robotics, IoT* — **August 2025**

- Built a low-cost (under $50) autonomous mapping robot: an HC-SR04 ultrasonic sensor on a servo performs 180° scans while a NodeMCU ESP8266 streams the resulting indoor obstacle map to a live web interface.
- Implemented polar-to-Cartesian coordinate conversion for real-time map plotting and closed-loop obstacle avoidance driven by an L298N motor driver, dual DC gear motors, and wheel encoders.

Hybrid Router: Two-Stage Local-First LLM Agent | *LLM Routing, Docker, Python* — **July 2026**

- Built a token-efficient LLM routing agent for AMD Hackathon ACT II (Track 1); a grammar-constrained local GGUF classifier (Qwen2.5-3B, Q4_K_M) resolves shallow tasks at zero API cost, escalating only high-difficulty queries across 8 intent categories to a Fireworks remote backend.
- Dockerized the full inference pipeline for a CPU-only grading VM (4 GB RAM, no GPU, linux/amd64) with env-driven model selection, 25s per-request timeout, graceful local fallback, and an offline pytest evaluation harness ensuring zero hardcoded values.

16-bit Binary Square Rooter | *VLSI, Verilog, SystemVerilog, Cadence EDA* — **Jan. – Apr. 2026**

- Designed a multiplier-free 16-bit integer square-root module in Verilog (shift-and-subtract, 8-iteration unrolled); achieved single-cycle throughput at 100 MHz and 100% functional coverage across directed, layered-SV, and covergroup testbenches.
- Executed full RTL-to-GDS flow (Cadence Genus and Innovus, 45 nm GPDK045): 197 cells, 347 sq. um area, DRC-clean layout with zero timing violations – setup WNS +0.264 ns, hold WNS +0.014 ns.

Brain Tumor Detection on MRI Images | *Digital Image Processing, MATLAB* **Mar. 2024**

- Developed a CAD system for automated brain tumor detection; improved MRI image clarity via advanced processing techniques.
- Achieved high sensitivity and specificity in tumor detection on clinical MRI data.

Research Interests

LLMs for Automated Hardware Design & EDA | AI Hardware and Edge Computing | Model Optimization and Efficient Inference | Embedded and Wearable AI | Robotics and Autonomous Systems | Computer Vision | Natural Language Processing

Relevant Coursework

Core Foundations: Linear Algebra | Probability and Statistics | Electrical Circuits | Energy Conversion | Electrical Service Design | Computer Programming

Signals and Communications: Continuous Signals and Linear Systems | Digital Signal Processing | Random Signals and Processes | Communication Systems | Wireless Communication | Radar and Satellite Communication | Optical Communications

Intelligent Systems and AI: AI and ML | Medical Image Processing | Microprocessors and Embedded Systems

Electronics and Hardware: Solid State Devices | Electronic Circuits | Digital Electronics | VLSI Circuits and Design | Power Electronics | Power System | Control System

Technical Skills

Languages: Python, SQL, C/C++, Verilog, SystemVerilog, HTML/CSS, Matlab, Bash, LaTeX
Frameworks and Libraries: PyTorch, TensorFlow, Keras, OpenCV, Hugging Face, Unsloth, llama.cpp, scikit-learn, Pandas, NumPy
Specializations: Large Language Models (LLMs), NLP, LLM Routing, Computer Vision, UAV, VLSI Design, RTL-to-GDS, PCB Design
Developer Tools: Docker, Cadence Genus, Cadence Innovus, Cadence Xcelium, Visual Studio, PyCharm, Kaggle, Colab, SolidWorks, LangChain, PSS/E, PSAF, Intel Quartus, Raspberry Pi, Arduino, Git, EasyEDA Pro

Volunteer Experience

BUET Robotics Society **March 2023 – June 2026**
Head of Design Team, Deputy Head of Event Management *Dhaka, Bangladesh*

- Led design and event teams, coordinating robotics projects and managing key events.

IEEE BUET Student Branch **January 2024 – June 2026**
Chairperson (SPS), Vice Chairperson (IAS), Executive Member *Dhaka, Bangladesh*

- Coordinated industry-academia collaborations and organized technical workshops and events.

Achievements

1st Runner Up (Int'l), IEEE-IAS Humanitarian Contest **IEEE IAS Annual Meeting 2025**

1st Runner Up, IEEE-WIE Robotics for Climate Change **WIE BD Summit 2024**

2nd Runner Up, Project Showcasing **Intra BUET Robo Challenge**